

# Grade 7 Mathematics — Mock Exam — Paper A

MEDIUM • NON-CALCULATOR • 75 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to work without one.
- **75 minutes for 80 marks** — about a minute a mark. If a 3-mark question has taken you five minutes, leave it and come back to it.
- No formulas and no methods are given anywhere on this paper — not one. Remembering the right one is part of what is being tested.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before you simplify.
- The mark for each question is shown in square brackets on the right.

## SECTION 1 – QUICK FIRE

1. Write down the answer to each of these.

(a) Write **84** as a product of its prime factors, in index form. [1]

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(b)  $-7 \times -6$  [1]

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(c)  $\sqrt[3]{216}$  [1]

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(d) Write  $3^4$  as an ordinary number. [1]

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(e)  $45 \times 0.1$  [1]

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2. Write down the answer to each of these.

**(a)** Round **6.472** to 1 decimal place. [1]

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**(b)**  $0.3 \times 0.4$  [1]

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**(c)** Write  $\frac{3}{8}$  as a decimal. [1]

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**(d)** Write the ratio 18 : 24 in its simplest form. [1]

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**(e)** Expand  $3(x + 5)$ . [1]

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SECTION 2 – SHORT ANSWERS

**3.** Two numbers are 12 and 18.

**(a)** Find the highest common factor (HCF) of 12 and 18. [1]

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**(b)** Find the lowest common multiple (LCM) of 12 and 18. [1]

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*One factor tree for each number answers both parts. Do not build them twice.*

**4.** Simplify  $(2^5 \times 2^3) \div 2^6$ . Give your answer as an ordinary number. [2]

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**5.** Work out:

(a)  $6.2 \div 0.1$

[1]

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(b)  $6.2 \times 0.01$

[1]

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*One of these answers is bigger than 6.2 and one is smaller. Decide which before you write anything.*

6. Write these decimals in ascending order — smallest first.

[2]

**0.7   0.68   0.702   0.71**

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7. Write these fractions in ascending order — smallest first. You must show your working.

[2]

$\frac{2}{3}$     $\frac{5}{8}$     $\frac{7}{12}$     $\frac{3}{4}$

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*Two of these four are close together. Looking at them will not separate them.*

8. Work out  $4\frac{1}{4} - 1\frac{5}{6}$ . Give your answer as a mixed number in its simplest form.

[2]

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9. A pencil costs ₹ $p$ . A pen costs ₹3 more than a pencil. Write an expression, in its simplest form, for the total cost of 4 pencils and 2 pens.

[2]

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10. Factorise fully:

**(a)**  $6x + 15$

[1]

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**(b)**  $8y^2 - 12y$

[1]

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*“Fully” means nothing is left inside the bracket that both terms still share.*

SECTION 3 – MULTI-STEP

**11.** Work out:

**(a)**  $-6 \times -4$

[1]

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**(b)**  $-6 - 4$

[1]

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**(c)**  $-48 \div 8$

[1]

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*(a) and (b) look almost the same and are not. Two minus signs multiplied make a plus; two minus signs added do not.*

**12.** Work out  $4.68 \div 0.12$ , showing your method.

[3]

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**13.** Work out  $6 \times 2\frac{3}{4}$ . Give your answer as a mixed number.

[3]

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**14.** Work out  $8 \div \frac{2}{5}$ . [3]

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*Dividing by a fraction smaller than 1 gives an answer bigger than 8.*

**15.** The cost, ₹ $C$ , of hiring a hall for  $n$  hours is given by  $C = 40n + 250$ .  
(a) Find  $C$  when  $n = 12$ . [2]

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(b) Find  $n$  when  $C = 1050$ . [1]

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**16.** Expand and simplify  $5(2x - 3) - 2(x - 4)$ . [3]

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*$-2 \times -4$  is plus 8. That one sign is where this question is usually lost.*

**17.** Solve  $5x + 7 = 3x + 19$ . Show every step. [3]

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**18.** A jacket costs ₹840. Its price is increased by 15%. Work out the new price. [3]

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SECTION 4 – PROBLEMS

**19.** ₹3600 is shared between Asha and Bela in the ratio 3 : 5.

**(a)** Work out how much each of them receives. [2]

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**(b)** Bela's share is then increased by 20%. Work out her new amount. [2]

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**(c)** How much more than Asha's share is Bela's new amount? [1]

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*Check your two answers to (a) add back up to 3600 before you go on.*

**20.** Ravi thinks of a number,  $n$ . He multiplies it by 4 and then subtracts 7. His answer is the same as adding 11 to the number he first thought of.

**(a)** Form an equation in  $n$ . [2]

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**(b)** Solve your equation, and check your answer works. [3]

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**21.** Two shops sell the same rice.

**Shop A:** 1.5 kg for ₹96. **Shop B:** 2.5 kg for ₹155.

**(a)** Work out the price per kilogram at each shop. [3]

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**(b)** Write down which shop is better value. [1]

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**(c)** Work out the cost of 6 kg at the better-value shop. [1]

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**22.** Two lights are switched on together at 9:00:00. One flashes every 12 seconds and the other flashes every 18 seconds.

**(a)** Work out how many seconds pass before they next flash together. [3]

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**(b)** Write down the time at which they next flash together. [1]

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**(c)** How many times do they flash together in the 6 minutes after 9:00:00? [1]

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**23.** A shirt costs ₹800. In one week its price is increased by 25%. The following week the new price is reduced by 20%.

**(a)** Work out the price after the increase. [2]

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**(b)** Work out the price after the reduction. [2]

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**(c)** Write down the overall percentage change from ₹800. [1]

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**24.** A rectangle has length  $(x + 4)$  cm and width 3 cm.

**(a)** Write an expression for its area. Expand your answer. [2]

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**(b)** Write an expression for its perimeter, in its simplest form. [2]

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**(c)** Factorise your expression for the perimeter. [1]

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*End of paper. Before you hand it in, check every answer that came out as a round, comfortable number — those are the ones a wrong method produces most often.*